

HELMET SENSE: AI-POWERED DETECTION OF HELMET RULE VIOLATION FOR ENHANCED ROAD SAFETY

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ABSTRACT:

The existing state of India's traffic laws presents several challenges, each of which can be approached in various ways. The rising number of fatalities related to motorbike and moped accidents has been attributed to the widespread practice of riding without a helmet in India. Currently, the system heavily relies on CCTV footage to monitor traffic violations, requiring law enforcement officers to manually review individual frames and zoom in on the numberplate if the rider is not wearing a helmet. However, this process demands a significant investment of time and resources, considering the prevalence of traffic offenses and the daily increase in the number of motorcyclists. What if there was a technology capable of detecting whether a motorcyclist is violating the law by not wearing a helmet and, if so, retrieving the vehicle's registration number from the database? Recent successful studies have achieved this using HaaR, HoG, LBP, R-CNN, CNN features, and other techniques. 1. Nevertheless, the speed, accuracy, and efficiency of object recognition and classification are significantly limited by these studies. To streamline the identification of instances where drivers fail to wear helmets and to capture vehicle number plates, this research introduces a Non-Helmet Driver identification technique. The fundamental concept involves employing three layers of Deep Learning for Image Classification. Each layer recognizes individuals, motorcycles, helmets, and license plates using YOLOv2's progressively advanced capabilities. Subsequently, Optical Character Recognition (OCR) is utilized to extract the license plate number. Specifically for license plate number extraction, all these techniques adhere to a predefined set of parameters and constraints. Given that video footage serves as the input, processing speed becomes crucial. Through the integration of the aforementioned methodologies, we have devised a comprehensive system for helmet identification and license plate number extraction.

INTRODUCTION:

The Globe Health Agency created a report called "The 2016 global study on traffic safety 2018" which reported that traffic accidents around the world cause over 3 million deaths and injure 1.5 million people every year. It's difficult to believe that pedestrians, cyclists, and drivers all bear equal responsibility for this issue.

Based on the conclusions drawn in this study, a comprehensive action plan is essential to reduce avoidable deaths. India holds the unfortunate distinction of having the highest number of fatalities from car accidents globally. Analysts have suggested that the surge in traffic-related deaths could be linked to various factors, such as the growing number of megacities and the general lack of adherence to basic safety measures like wearing helmets or seatbelts while driving. In 2015, India committed to the Madrid Statement on Traffic Safety, aiming to halve the number of traffic accident fatalities by 2020.

To reduce the number of fatalities in road accidents in India, authorities must acknowledge the existing issues. When a motorcycle or scooter rider is involved in a collision, the sudden deceleration from the impact often leads to ejection from the vehicle. Even if the movement of the

neck halts post-impact, the body's momentum causes it to continue moving until it strikes the interior of the skull. In some instances, a severe brain injury could prove fatal. Wearing a helmet can be a lifesaving measure in such scenarios. By reducing the chances of the skull coming to an abrupt stop, a helmet effectively prevents the head from moving.

In case of a collision, the internal cushion of the helmet absorbs the majority of the impact and gradually brings the head to a stop. Additionally, it disperses the force over a wider area, safeguarding the skull from any potential harm. In the event of an accident, a high-quality full headgear can potentially prevent or reduce the severity of head injuries by serving as a physical barrier between the rider's head and any object the rider collides with.

The purpose of traffic laws is to promote professionalism and minimize the risk of accidents and resulting injuries. Nevertheless, in reality, these regulations are not adhered to.

It is imperative to devise efficient and feasible methods to address this issue. The utilization of closed-circuit television for manual traffic monitoring has emerged as a viable solution. Nevertheless, this approach necessitates numerous repetitions to achieve the desired outcome. Consequently, cities with a substantial population and heavy traffic cannot rely on this inefficient manual technique for identifying helmets. To overcome this challenge, we present a comprehensive helmet recognition and license plate retrieval technique using OCR, YOLOv3, and YOLOv2.

1. YOLO serves as a real-time object identification system capable of recognizing objects in both still images and moving videos. An enhanced and more accurate iteration of the YOLO algorithm, known as YOLO v2, has been developed. Optical Character Recognition (OCR) can be utilized to extract text from images and videos, with one potential application being the extraction of license plate numbers from photographs.

The combination of YOLO v2 and optical character recognition enables the identification of motorcyclists without helmets and extraction of their licence plate numbers. By utilizing the YOLO v2 algorithm, pictures or videos containing motorcyclists can be analyzed, while optical character recognition can accurately read the licence plate numbers. This collected data can then be utilized to identify and apprehend individuals who violate helmet regulations.

LITERATURE REVIEW:

A. First, we have Headgear Recognition.

DL has recently witnessed numerous new proposals. In [1], thresholding and the MOC decoder are discussed as techniques for detecting motorbikes in videos. CNN is employed with manually crafted features to determine whether individuals are wearing helmets or not. CNN outperforms human-created features in this regard. To extract a moving object from a video, adaptive local removal is utilized. The CNN technique is used to categorize motorcycles into different vehicle classes. Additionally, CNN is used to inspect the heads of motorcyclists and ensure that they are not wearing protective headgear [3]. The GMM is employed to isolate and label the surrounding elements. A faster Geographical area Convolutional Neural Network is employed to specifically search for motorcycles in the background and confirm their presence. The R-CNN is then utilized to identify motorcyclists who are wearing or not wearing protective headgear. Although D-CNN networks are used for headgear identification [1,2,3], the background object in the motorcycle initiation phase is still obtained using a traditional median filter, which may not perform well in a high-noise environment.

The identification of motorcycles remains unconfirmed, although it is suggested in references [4] and [5] that the YOLOv3 technique is used to identify bikers wearing helmets. Initially, the YOLOv3

method was applied to locate the motorcycle with a rider in the images from [7] and [8]. Subsequently, an evaluation of the overlapping grid points containing both objects was conducted to determine the rider. The YOLOv3 techniques were then utilized to verify if the bikers were wearing helmets. Therefore, precise detection of motorcyclists is deemed unnecessary once transportation has been observed due to the significant overlap between the two categories. The proposal in references [9, 10, 11] involved applying SSD/YOLOv3 techniques to identify the motorcycle area, cropping out the head, and using a classifier to differentiate between helmets and no helmets. However, this method may not be effective if there are multiple riders on the motorcycle. In [12], [13], and [14], the CNN architecture was employed to determine whether the rider was wearing a helmet, taking into account both the motorcycles and the riders. These are single-step, high-precision detection methods.

B. Plate Readers

Numerous researchers have explored the potential of utilizing convolutional neural networks (CNNs) for object recognition in the LP detection process. According to Silva&Jung [15], the Quick algorithm [16] demonstrated limited success in accurately identifying LPs without first detecting the vehicles. A cascaded approach involving Fast-YOLO was implemented to first recognize the top view of the vehicles and subsequently their LPs at specified positions, resulting in high levels of accuracy recall for a dataset containing Brazilian LPs.

HSU[17] modified the YOLO and YOLOv2 algorithms specifically for LP detection. It is essential for LP detection methods to be faster than the updated YOLO, which currently processes 54 FPS with a decent GPU, as the LP elements still require accurate recognition. On the other hand, Kurpiel[18] developed a nested, overlapping grid using image data. Scores were computed for each grid using a CNN, and LPs were identified by examining the results in neighboring sub-regions. Their real-world data demonstrates that identifying Brazilian LPs in images containing multiple car parts takes only 230 milliseconds on a GT-740M GPU, achieving an 83% reliability rate.

Li[19] utilized letters extracted from various texts to train a CNN for character-based LP detection. By rolling a frame across the entire image, the networks were able to create a text salience map. Clusters of characters were then utilized to identify the locations of text segments. Subsequently, causal research was employed to establish the initial criteria and selection boxes. Following this, a convolutional neural network (CNN) with LP and non-LP layers was trained to eliminate FPs. Nevertheless, the cost associated with using this array of techniques makes them unsuitable for practical applications; for instance, when considering an Electric car or selecting a suitable Graphics card, processing a simple photograph can take over 2 seconds, rendering them impractical even if the precision and recall rates achieved were superior to those obtained in previous studies.

C. The YOLO Era

The YOLO model, introduced in 2015 by JosephR in the work "YOLO: Unified, Context Of physical Identification," revolutionized object recognition by enabling it to be done in a single pass. Prior to YOLO, RCNN models were the go-to for object identification, despite being effective, they were slow due to the multiple steps involved such as identifying proposed regions, categorizing those areas, and post-processing to refine the results. YOLO was designed to streamline the detection process into a single phase, reducing inference time while increasing computation time.

METHODOLOGY:

This section discusses various stages of the process. As shown in Figure 1(a) and Figure 1(b), the initial step involves capturing images from the video stream at regular intervals. Each shot is stored in a separate directory. The filename includes the frame count, such as frame3-10, frame3-20, and so on. The numbers 10, 20, etc. represent the frames in the film or where the 3rd video file was inserted. The data clearly indicates that many of the frames are unnecessary. In such cases, the final image or the second-to-last frame is selected based on the data regarding the relative motion of the vehicle and the cameras.

For two different scenarios, the entire process can be divided into five stages:

Scenario 1: Motorcyclist wearing a helmet

Scenario 2: Motorcyclist not wearing a helmet.

1. DETECTION OF MOTORCYCLES AND RIDER

The chosen image is subsequently inputted into a YOLOv2 model for identification and verification, with "Motorcycle" and "Biker" as the pre-defined categories. As illustrated in Figure 2 (a) and Figure 2 (b), the end result is an image that includes accurate class detection, along with a high level of confidence in detection across consecutive frames and the significance level.



2. DETECTION OF HELMETS

After establishing the user combination, the helmet classification algorithm was applied to the photos of the people. The headgear detection algorithm identified false positives. To address this issue, the top quarter of the original human picture in Fig. 4 was cropped, effectively removing the false identification and erroneous results. This cropping technique prevents occurrences where the driver is not actually wearing the headgear but instead holding it or leaving it on their motorbike. The headgear recognition technique was then implemented on the cropped picture.



The results are illustrated in Fig. 5, showcasing the helmet's recognition probability and its corresponding reference image. In Scenario 1, where the rider wore a safety helmet, there is no need for any further action. Conversely, in Scenario 2, where no headgear is present, there is no requirement to generate a limiting arra.

3. NUMBER PLATE RECOGNITION FOR VEHICLES

If the protective headgear is found, you can proceed to the next step. In case the headgear is not found, the motorcycle image will be used as input in a license plate classification algorithm. A total of 832 photos of bicycle and moped license plates were collected for machine learning purposes. These photos were labeled using a labeling tool to create a bounding box around each plate for the model to analyze. The information is then recorded in a separate XML file with the same filename as the picture. Subsequently, the labeled images are used to generate the training set for license plate detection.



A specific image source is employed by the learning algorithm to create a boundary around the license plate. The dimensions of the bottom-right and top-left subsets, class label, and certainty are all specified in the corresponding JSON file. Subsequently, the JSON file's bounding values are used to extract the license plate image selectively. In Figure 6, multiple clusters are identified within the same motorcycle image. The detection threshold is then established at 0.5. The JSON file includes information about the structuring element, and the one with a score exceeding the threshold is chosen.

Prior to utilizing OCR on the captured license plate image, it is essential to ensure that the process is carried out in order to achieve higher accuracy. Consequently, the image was rotated. Figure 7 illustrates the appearance of the license plate picture after it has been cropped and rotated.

The degree by which the retrieved license plate picture must be flipped will be determined experimentally only once, as the camera will always remain in the same relative position to the motorbike. After this initial determination, the same degree of 6° will be used for all future instances.

The character detection rate of OCR was enhanced by resampling the rotated image. The resizing of the image was achieved by choosing a scale ratio, which represents the width to body ratio of the resized image compared to the original one. Let the initial picture's width and height be denoted as w and h , respectively, and the resized image's height and width as w' and h' . Let's denote r as a proportion for illustration purposes.

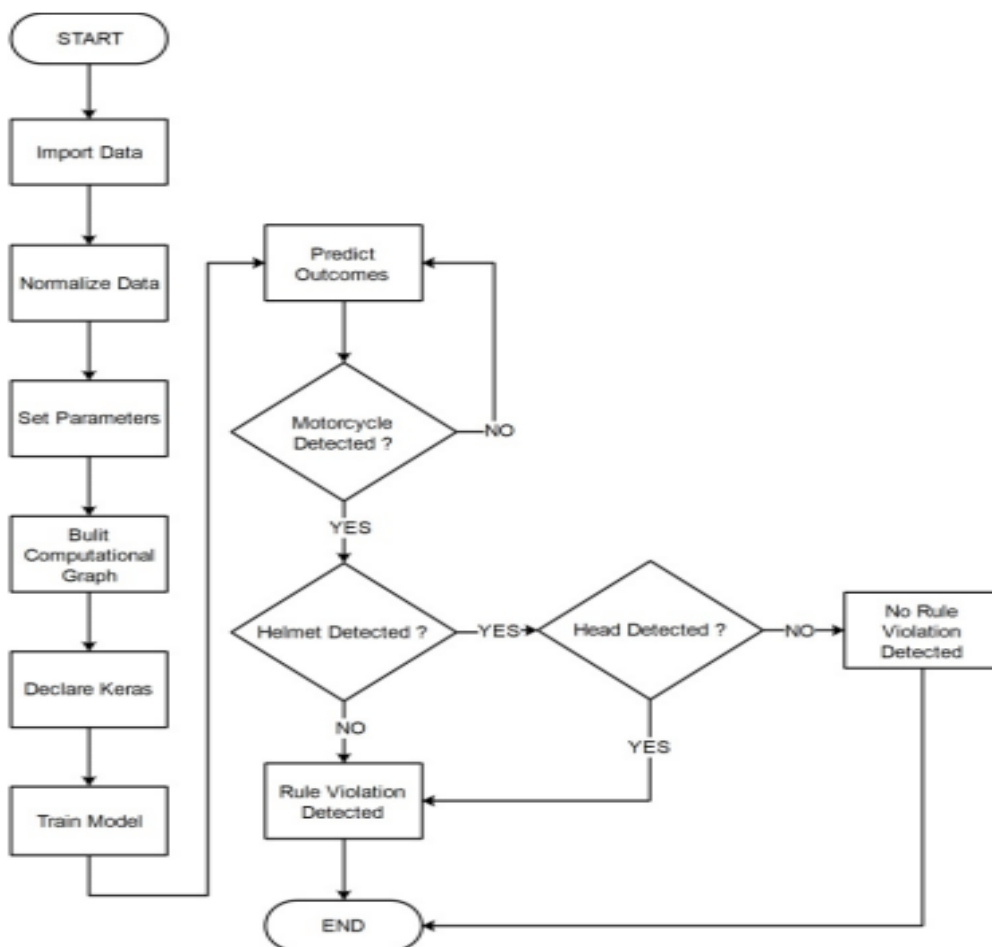
Then the rescaled image size is obtained by:

$$w = wr$$

$$h = hr. \quad (2)$$

The dynamic ratio, denoted as ' r ', undergoes changes depending on the selected frame during frames separation. The findings revealed that the specific value fell within the range of 1.4% and 1.5%. Subsequently, the contrast of the image is enhanced to make the black license plate numerals more distinct against the white background. The image's Color, Intensity, and Brightness (h,s,v) values were measured. It is widely recognized that the ' v ' (Value) of a color indicates its lightness or darkness. For pixels with a ' v ' value exceeding a certain maximum, 226 has been assigned as the value for that pixel. Conversely, if the ' v ' value of a pixel is lower than the minimum, a predetermined value has been added to it. In this example, the maximum value is 226, while the equilibrium point is set at 30. The value is limited to 255 if ' v ' exceeds the limit, otherwise, ' v ' is incremented by the value. This serves as an illustrative example of a figure caption.

FLOW CHART:-



RESULTS :-

In this article, we go through the outcomes for two different scenarios. It's true that they are, To begin, consider the situation depicted in fig

Fig. shows what happens in Case 2 when a motorbike rider is not using a helmet...

Table 1. Details of Threshold value with model

Sl.no	Detection Model	Number Plate Detection	Threshold value
1	YOLO v2(Without Helmet)	YES	0.5
2	YOLO v2(with Helmet)	NO	0.87

CONCLUSION:

The YOLO image identification architecture is utilized in the development of a system that detects riders without helmets using a video clip as input. If the driver in the video clip is not wearing a helmet, the license plate of the motorbike is retrieved and displayed. Both the motorbike and the individual's headgear are recognized using the same architecture. In order to extract the license plate number from the vehicle, OCR technology is employed when the rider is not using a helmet. The characters from the frames are extracted, along with the structure itself. The project has achieved its goals successfully.

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